Name: Sumit Chauhan

Dept: Data Engineering

Module: ADF: Key Features, Use Cases, Key Terms

Task:

Create Linked Services to connect Azure Data Factory with Azure Data Lake Storage (ADLS) and Microsoft SQL Server database. Configure necessary authentication details (e.g., Azure Key Vault for secrets).

Create datasets for both the source (CSV files in ADLS) and the destination (tables in SQL Server).

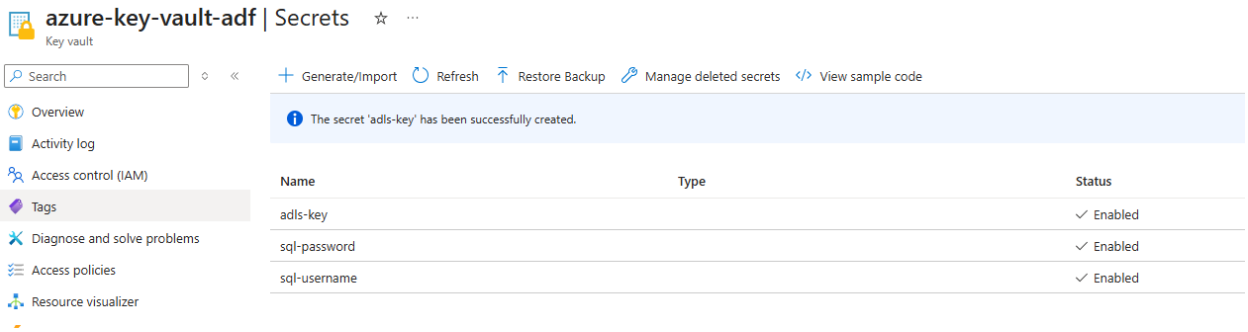
Create a pipeline to copy data from ADLS to SQL Server and vice versa. Use a copy activity to move files between a folder in ADLS and a table in SQL Server.

Initially create the following resources in Azure:

ADF, SQL database and blob container (in storage account)

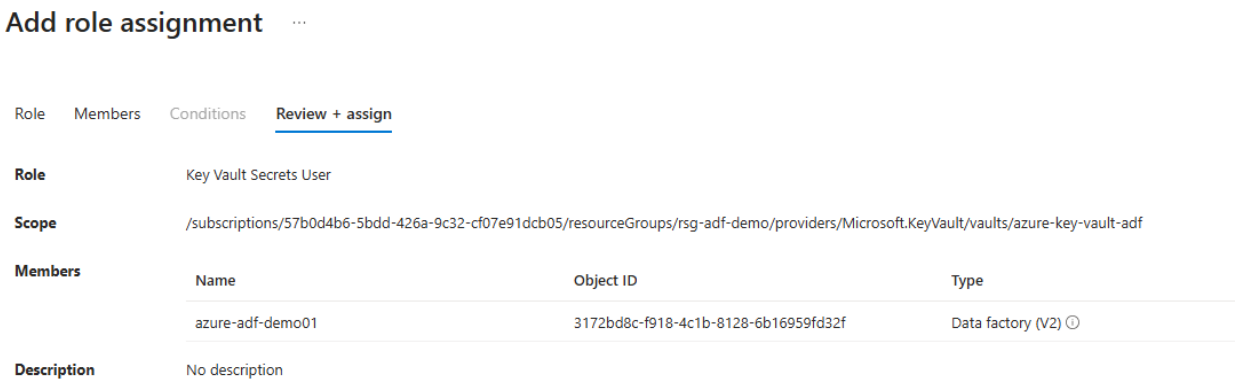
Create key-vault and store necessary credentials

- Storage account connection string → `adls-key`
- SQL username → `sql-username`
- SQL password → `sql-password`



Add role assignment to ADF (as managed identity) from key-vault so that it can access the secrets

Make sure that you store connection-string in adls-key instead of access-key



Add linked services for blob and sql database:

And for secure credential management, create linked service for azure key vault that connects with key-vault and use the secrets

Edit linked service

Azure Key Vault

Name *

AzureKeyVault1

Description

Azure key vault selection method

From Azure subscription Enter manually

Base URL *

https://azure-key-vault-adf.vault.azure.net/

Authentication method

System-assigned managed identity

Managed identity name: **azure-adf-demo01**
Managed identity object ID: **3172bd8c-f918-4c1b-8128-6b16959fd32f**
Grant Data Factory service managed identity access to your Azure Key Vault. Learn more

Test connection

To linked service To secret

Annotations

+ New

> **Parameters**

> **Advanced**

Save Cancel

Connection successful

Test connection

Edit linked service

Azure Blob Storage Learn more

Name *

inputBlob

Description

Connect via integration runtime *

AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string | Azure Key Vault

AKV linked service *

AzureKeyVault1

Secret name *

adls-key

Edit

Secret version

Latest version

Edit

Test connection

To linked service | To file path

Annotations

New

Parameters

New | Delete

Connection successful

Test connection

Save | Cancel

Edit linked service

Azure SQL Database Learn more

Name *
outputSQL

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Version
2.0 (Recommended) 1.0

Account selection method
From Azure subscription Enter manually

Fully qualified domain name *
azure-sql-server-adf.database.windows.net

Database name *
azure-sql-db-adf

Authentication type *
SQL authentication

User name *
sqladmin

Password Azure Key Vault

AKV linked service *
AzureKeyVault1

Secret name *
sql-password

Edit

Connection successful

Save Cancel Test connection

Linked services

Linked service defines the connection information to a data store or compute. Learn more

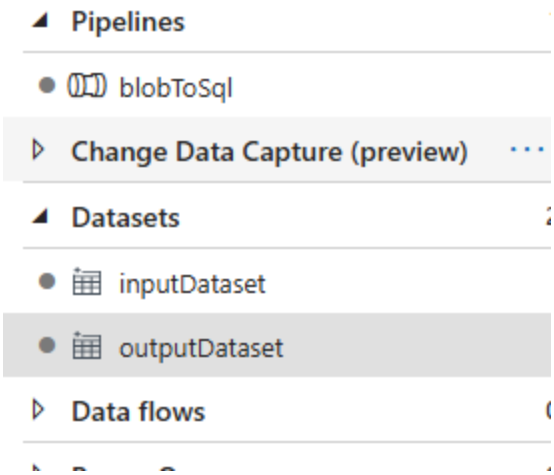
+ New

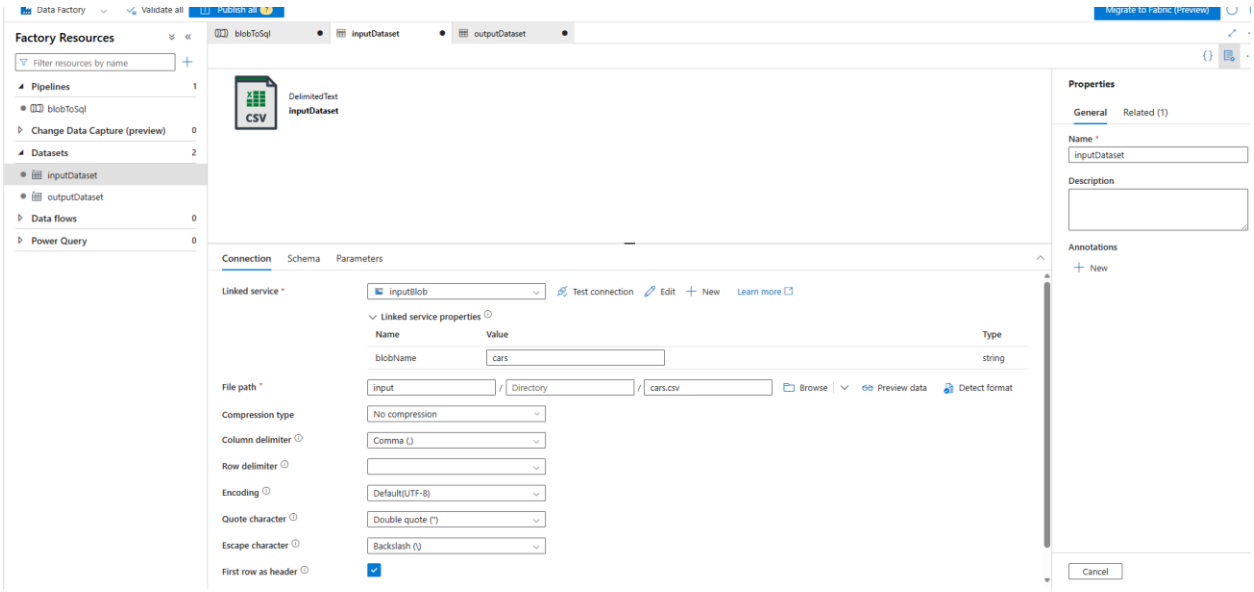
Filter by name Annotations : Any

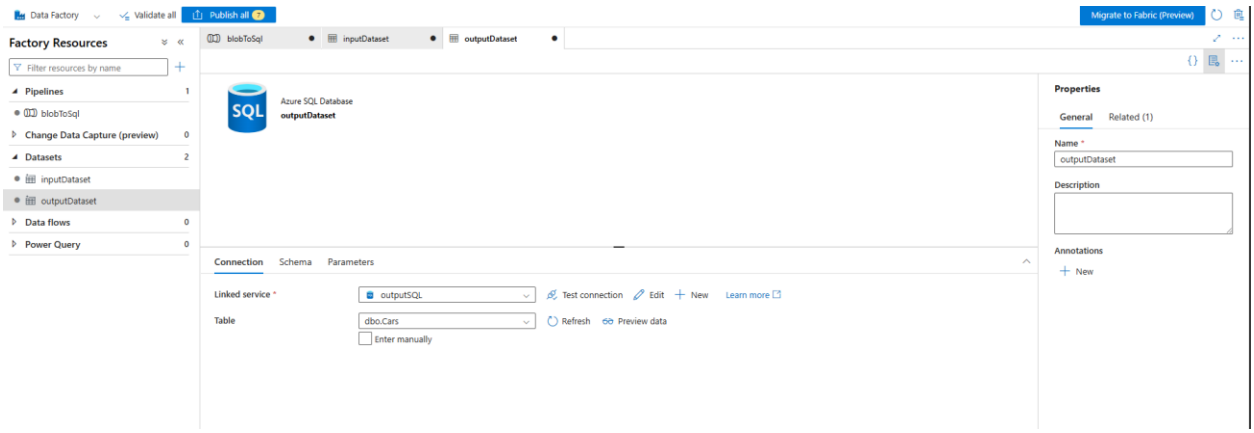
Showing 1 - 3 of 3 items

Name	Type	Related
AzureKeyVault1	Azure Key Vault	2
inputBlob	Azure Blob Storage	1
outputSQL	Azure SQL Database	1

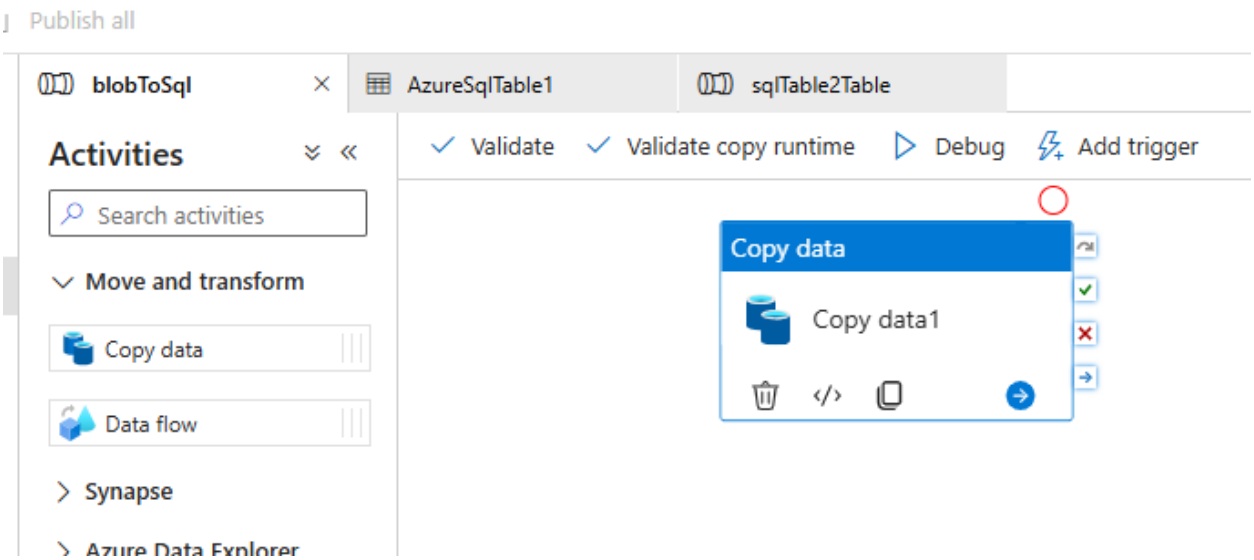
Then add the datasets in pipeline







Then create a pipeline from Author's tab from ADF



You add different types of activities in it

For our use case we have added Copy data activity which will copy the blob storage data to sql database table